Experiment 8

Aim: Study and implement Database as a service using AWS RDS and DynamoDB

Theory:

1. Study Database as a service.

Database as a Service (DBaaS) is a cloud computing service that provides database functionality without the need to set up physical hardware or manage complex database software. In DBaaS, the cloud provider manages the infrastructure, database installation, updates, backups, scaling, and security.

Users only need to create and configure the database and can access it through the internet. This allows organizations to focus on application development rather than database maintenance.

In DBaaS:

- The database runs on cloud servers.
- Users access it through web interfaces or APIs.
- Maintenance tasks like backups, patching, and scaling are automated.

Examples of DBaaS include:

- Amazon Web Services services like Amazon RDS and Amazon DynamoDB.

Advantages:

- Reduced infrastructure cost
- High scalability
- Automatic backups and recovery
- Easy accessibility from anywhere

2. Why is a database important?

A **database** is a structured collection of data that allows efficient storage, retrieval, and management of information.

Importance of databases:

1. Organized Data Storage
Databases store large amounts of data in an organized structure.

2. Efficient Data Retrieval
Data can be quickly searched and retrieved using queries.

3. Data Integrity
Databases ensure that stored data remains accurate and consistent.

4. Security
Access controls protect sensitive information.

5. Data Sharing
Multiple users or applications can access the same data simultaneously.

6. Backup and Recovery
Databases allow safe storage with backup mechanisms to prevent data loss.

Applications such as banking systems, social media platforms, and e-commerce websites rely heavily on databases to operate efficiently.

3. What are modern Databases?

Modern databases are advanced database systems designed to handle large-scale data, high-speed processing, and cloud-based environments.

Unlike traditional databases, modern databases support:

- Large volumes of data (Big Data)
- Real-time processing
- Cloud integration
- High scalability
- Distributed computing

Types of modern databases include:

- Relational Databases
- NoSQL Databases
- Graph Databases
- Document Databases
- Cloud Databases

Examples include:

- MongoDB
- Cassandra
- Amazon DynamoDB
- MySQL running on cloud platforms.

These databases are designed to handle modern applications such as social networks, real-time analytics, and large-scale web services.

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4. What is a Relational Database?

A **Relational Database** is a type of database that stores data in the form of tables consisting of rows and columns.

Each table represents an entity, and relationships between tables are established using keys such as:

- Primary Key
- Foreign Key

Characteristics:

- Structured schema
- Uses **Structured Query Language (SQL)** for data manipulation
- Maintains data integrity using constraints
- Supports complex queries and joins

Example relational database systems include:

- MySQL
- PostgreSQL
- Oracle Database

In cloud environments, relational databases are commonly deployed using services like Amazon RDS.

5. Explain Amazon DynamoDB and Features and characteristics.

Amazon DynamoDB is a **fully managed NoSQL database service** provided by Amazon Web Services. It is designed for applications that require fast and predictable performance with seamless scalability.

Unlike relational databases, DynamoDB stores data in **key-value pairs or document format**.

Features

1. Fully Managed Service
AWS manages hardware, scaling, and maintenance.

2. High Performance
Provides single-digit millisecond latency for data access.

3. Automatic Scaling
Capacity automatically adjusts according to application demand.

4. High Availability
Data is automatically replicated across multiple servers.

5. Security
Supports encryption and access control through AWS IAM.

6. Flexible Data Model
Supports both key-value and document data models.

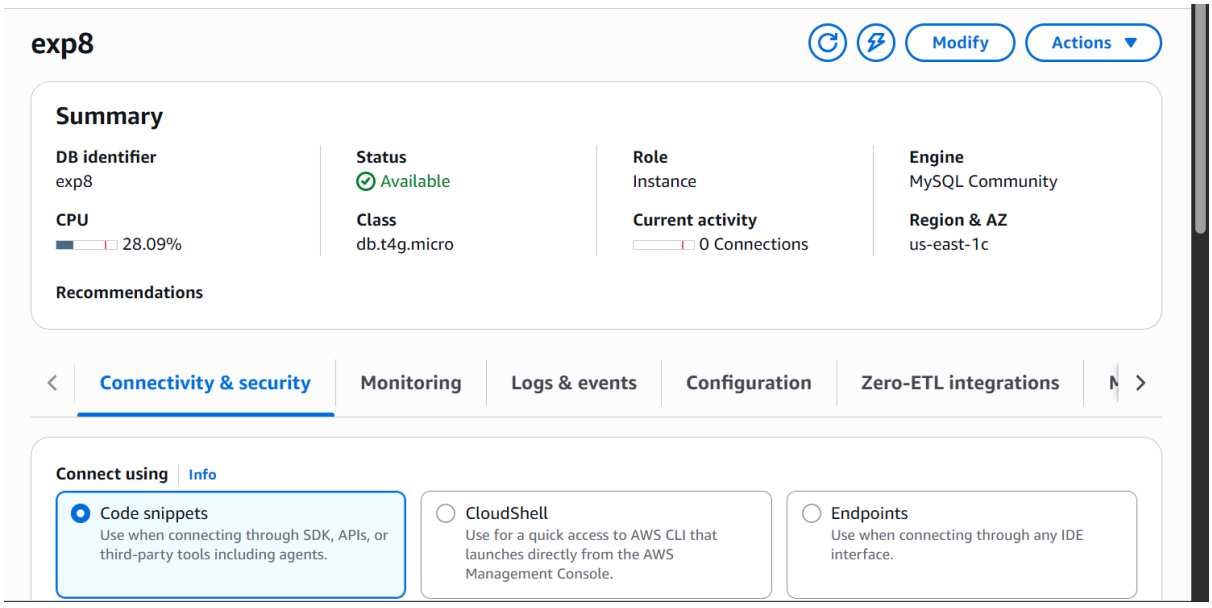
Characteristics

- Serverless architecture
- No schema restrictions
- Highly scalable
- Fault tolerant
- Suitable for real-time applications

DynamoDB is commonly used for:

- Gaming applications
- Mobile apps
- IoT systems
- Real-time analytics platforms

ACTIVITY:



Experiment 8

```
~ $ mysql -h exp8.cube4s0gokwp.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 26
~ $ mysql -h exp8.cube4s0gokwp.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 26
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE college;
Query OK, 1 row affected (0.074 sec)

MySQL [(none)]> USE college;
Database changed
MySQL [college]> CREATE TABLE student (
    ->     name VARCHAR(50),
    ->     roll_number INT
    -> );
Query OK, 0 rows affected (0.075 sec)
```

```
MySQL [college]> INSERT INTO student (name, roll_number)
    -> VALUES
    -> ('raghav', 1),
    -> ('dhaval', 2),
    -> ('shahiil', 3);
Query OK, 3 rows affected (0.059 sec)
Records: 3  Duplicates: 0  Warnings: 0

MySQL [college]> SELECT * FROM student;
+---------+-------------+
| name    | roll_number |
+---------+-------------+
| raghav  |           1 |
| dhaval  |           2 |
| shahiil |           3 |
+---------+-------------+
3 rows in set (0.002 sec)

MySQL [college]> UPDATE student
    -> SET roll_number = 66
    -> WHERE name = 'raghav';
Query OK, 1 row affected (0.013 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

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```
MySQL [college]> UPDATE student
    -> SET roll_number = 70
    -> WHERE name = 'dhaval';
Query OK, 1 row affected (0.042 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MySQL [college]> UPDATE student
    -> SET roll_number = 65
    -> WHERE name = 'shahiil';
Query OK, 1 row affected (0.008 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MySQL [college]> SELECT * FROM student;
+---------+-------------+
| name    | roll_number |
+---------+-------------+
| raghav  |          66 |
| dhaval  |          70 |
| shahiil |          65 |
+---------+-------------+
3 rows in set (0.024 sec)
```

Conclusion:

Compare different types of database technologies with aws:

In this experiment, the concept of **Database as a Service (DBaaS)** was studied and implemented using services provided by Amazon Web Services. Cloud platforms offer different types of database technologies to support various application requirements.

Relational databases store structured data in tables and support complex queries using SQL. Services such as Amazon RDS allow users to deploy relational databases like MySQL and PostgreSQL without managing hardware, backups, or maintenance. These databases are suitable for applications that require strong data consistency and structured relationships.

In contrast, NoSQL databases such as Amazon DynamoDB are designed for high scalability, fast performance, and flexible data models. They handle large volumes of unstructured or semi-structured data and are widely used in real-time web and mobile applications.

Thus, AWS provides a variety of database technologies that enable organizations to choose the most appropriate solution based on their data structure, scalability requirements, and performance needs. Cloud-based databases simplify management, improve reliability, and support modern application development.